B. Tech. Third Semester (Information Technology)/(SOE_23_IT-101) Examination

Course Code : 23 IT 1301 **Course Name : Data Sturcture and Program Design**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number. 391872

(2) Write the accurate question number in left margin of answer book along with answers
(3) All questions are compulsory.
(4) All questions carry marks as indicated.
(5) Due credit will be given to neatness and adequate dimensions.
(7) Illustrate your answers wherever necessary with the help of neat sketches.

1. Solve any Two :

(a) Write a program to implement a function to find the second largest element in an array without sorting. 4(CO1)

(b) Consider following program with the goal of the program is to get a word from the user and print out the word so that each letter is printed multiple times :

```
#include <stdio.h>
#include <string.h>
int main(void)
{
while (1)
{
printf("Enter a word or q to quit : ") ;
char word[51] ;
scanf("%50s", word) ;
if (strcmp(word, "q") = = 0)
```

Contd...

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```
break ;
int times = getTimes() ;
repeatLetters(word, times) ;
}
return 0 ;
}
```

Complete the program by defining two functions :

(1) A function called repeatletters that satisfies the following specs :

(a) Repeattetters takes two arguments, called text and times.

(b) The function prints each letter of text as many times as specified by times.

(2) A function called getTimes that satisfies the following specs :

(a) Asks the user for an integer greater than 0.

(b) If the user enters an integer less than 1, prints an error message and keeps asking them until they enter a valid integer.

(c) Returns the valid integer. 4(CO1)

(c) Create a COVID vaccine registration form using structure. Users will register by providing the following details :

First Name
Last Name
Dose Number (1 or 2)
Type of Vaccine (Pfizer, Moderna, Johnson & Johnson)
Residential Zipcode

The program must generate a unique 6-character alphanumeric code in the format :

<First letter of First Name> <First letter of Last Name> <First letter of Vaccine Type> <Last 3 digits of Zipcode>

Write c program which accepts data for 3 users and Display each user's information after registration, one item per line. 4(CO1)

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2. Solve any Two :

(a) Design an Abstract Data Type (ADT) for a Library System. The ADT should support operations such as adding a book, removing a book, searching for a book by title and listing all available books. Write the ADT, specifying the operations, their inputs and their outputs without detailing the implementation. Consider edge cases like searching for a non-existent book or adding duplicates. 4(CO2)

(b) Consider the following C function :

```
void printPairs(int arr[], int n) {
  for (int i = 0 ; i < n; i++) {
    for (int j = i + 1 ; j < n ; j++) {
      printf("(%d, %d) \ n", arr[i], arr[j]) ;
    }
  }
}
```

Calculate time and space complexity of above code. 4(CO2)

(c) Analyze the time complexity of this function in the best case, worst case, and average case scenarios for following code.

```
int linearSearch (int arr[], int n target) {
 for (int i = 0: i < n: i++){
   if (arr(i)= =target){
     return i;// Return the index if the target is found
   }
 }
return –1;// Return–1 if the target is not found
}
```

4(CO2)

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3. Solve any Two :

(a) Write a function to search some value in a stack. Function returns –1 if values is not in stack otherwise it returns how far that value is from top. 4(CO3,4)

(b) Convert following infix expression to postfix, show the status of stack in each step –

(A + B * C / D – E + F / G / (H + I)) 4(CO3,4)

(c) Consider the following queue, where queue is a circular queue having 6 memory cells. Front = 2, Rear = 4.

Queue : _ , A, C, D, _ , _

Describe queue as following operation take place :

F is added to the queue

Two letters are deleted

R is added to the queue

S is added to the queue

One letter is deleted. 4(CO3,4)

4. Solve any Two :

(a) Let list1 and list2 are two singly linked lists having data in ascending order. Write a function to merge them such that resultant list will contain data in ascending order. 4.5(CO3,4)

(b) Let assume doubly linked list, Implement a doubly linked list with functions to insert and delete elements at both ends. 4.5(CO3,4)

(c) Write a program to find the middle element of a singly linked list where elements are randomly arranged. 4.5(CO3,4)

5. Solve any Two :

(a) Construct B tree of order 5 for the following data :

1, 6, 7, 2, 11, 5, 10, 13, 12, 20, 16, 24, 3, 4, 18, 19, 14, 25 4.5(CO3,4)

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(b) Construct AVL tree for the following data. Mention the type of rotation for each case :

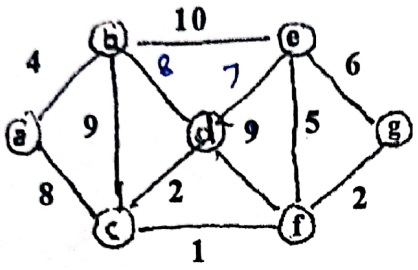
50, 25, 10, 5, 7, 3, 30, 20, 8, 15 4.5(CO3,4)

(c) Construct binary search tree for the following data and find its Inorder, Preorder and Postorder traversal :

10, 3, 15, 22, 6, 45, 65, 23, 78, 34, 5 4.5(CO3,4)

6. Solve any Two :

(a) Write and explain Kruskal's algorithm to find Minimum Spanning Tree (MST). Find MST for the following graph.



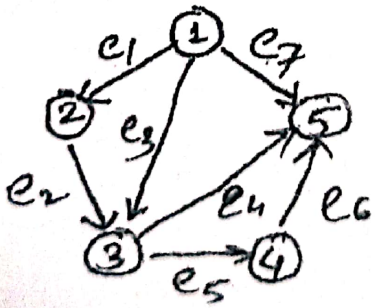
4(CO3,4)

(b) Assume that you construct a hash table using the separate chaining for the following 7 keys :

31, 72, 20, 14, 9, 2, 32

Assume also that the hash function is h(K) = K mod 5. For the problem, do not consider the rehashing. Draw the hash table after inserting all numbers. 4(CO3,4)

(c) Compare the adjacency matrix and adjacency lists representations used to represent graph in memory. Also state their advantages and disadvantages.



4(CO3,4)

B. Tech. Third Semester (Information Technology) / (SOE_23_IT-101) Examination

Course Code : 23 IT 1303 **Course Name : Computer Architecture and Organization**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number. 391872

(2) Write the accurate question number in left margin of answer book along with answers.

(3) All questions are compulsory.

(4) All questions carry marks as indicated.

1. Solve the following (any Two) :

(A) Discuss the following terms :

- MAR
- MDR
- Instruction Register
- Program counter

Show the connection between memory and processor. 4(CO1)

(B) Differentiate between computer architecture and organization. Also discuss the functional blocks of a computer. 4(CO1)

(C) Write a program that can evaluate the expression : A x B = C x D, in a single accumulator processor. Assume that the processor has load, store, multiply and Add instructions and that all values fit in the accumulator. 4(CO1)

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2. Solve the following (any Two) :

(A) Explain the role of registers transfer language in a CPU's instruction set architecture. 4(CO2)

(B) What is the significance of a bus organization in a computer system ? Write control sequence for instruction ADD (r2), r3. Register r2 contains memory address. 4(CO2)

(C) Discuss in detail how an instruction is executed with conditional branching with the help of neat diagram. 4(CO2)

3. Solve the following (any Two) :

(A) Differentiate between hardwired control and microprogrammed control. 4(CO1)

(B) How to reduce the length of microinstruction ? Explain with example. 4(CO1)

(C) Describe the significance of microinstructions with a next address field. 4(CO1)

4. Solve the following (any Two) :

(A) Obtain resultant from division of 10/3 using non restoring algorithm. 4(CO3)

(B) Using Booth's algorithm determine the product of (13 x –12). 4(CO3)

(C) Represent the following in single and double precision IEEE format for following floating point representation : -1.75×10^{15}. 4(CO3)

5. Solve the following (any Two) :

(A) Calculate set, tag and word bits for 4-way associated cache memory. Cache is having 128 blocks of 16 words each and size of main memory = 4095 blocks. 4.5(CO3)

(B) Discuss the following terms with respect to cache memory :—

(i) Locality of reference.

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(ii) Dirty bit.

(iii) Replacement algorithms.

(iv) Cache Miss. 4.5(CO3)

(C) Reference string is 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3. Determine number of page faults in the following replacement algorithms:

(a) Least Recently used replacement algorithm.

(b) Optimal Page Replacement algorithm.

(c) Determine which method yields less number of page faults. 4.5(CO3)

6. Solve the following (any **Two**) :

(A) Explain the use of Direct Memory Access (DMA) controller in a computer system. Also state about the registers used in DMA. 4.5(CO4)

(B) Define Interrupt. Elaborate in detail the concept of interrupt service routine in processor. 4.5(CO4)

(C) Write short note on instruction pipelining in computer system. Identify the type of Hazard in the following set of instructions :

ADD R1, R2, R3

SUB R4, R5, R1

AND R6, R7, R4 4.5(CO4)

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B. Tech. Third Semester (Civil Engineering) / (EL/ET/IT/AIDS/AIML) (SOE_23_CV-101) Examination

Course Code : 23 CV 1311 **Course Name : Environmental Sustainability, Pollution and Management**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number. 351872

(2) Write the accurate question number in left margin of answer book along with answers.

(3) All questions are compulsory.

(4) All questions carry marks as indicated.

(5) Due credit will be given to neatness and adequate dimensions.

1. Solve any **Two** :

(A1) Discuss the importance Land and Energy as a natural resource. 6(CO1)

(A2) Explain the concept of carrying capacity of an ecosystem. Define Sustainable Development, its pillars and enlist SD goals. 6(CO2)

(A3) Describe the impact of climate change on ecosystem. Discuss the mitigation measures to combat Climate Change. 6(CO3)

2. **Solve any Two** :

(A1) Discuss in brief about air pollution, classification and ill effects of air pollutants on ecosystems. 6(CO2,3)

(A2) Discuss in detail about noise as a pollutant. Discuss the consequences of soil pollution. 6(CO2,3)

(A3) Explain how environmental pollution affects the natural ecosystems. 6(CO2,3)

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3. Solve any **Two** :

(A1) Describe in detail about Environmental Impact Assessment. 6.5(CO1)

(A2) Discuss in detail about Environmental Management systems. 6.5(CO2)

(A3) Discuss important stages of Life cycle Analysis. 6.5(CO2)

4. Solve any **Two** :

(A1) Discuss in detail about E-Waste Management Rules, 2016 and Kyoto protocol. 6.5(CO4)

(A2) Discuss in brief about Paris agreement and Solid Waste Management Rules, 2016. 6.5(CO4)

(A3) Discuss the salient features of The Air Act, 1986 and Plastic Waste Management Rules, 2016. 6.5(CO4)

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B. Tech. Third Semester (Information Technology) (SOE_23_IT-101) Examination

Course Code : 23 IT 1304 **Course Name : Object Oriented Programming**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number. 391872

(2) Write the accurate question number in left margin of answer book along with answers.

(3) All questions carry marks as indicated.

(4) Due credit will be given to neatness and adequate dimensions.

1. **Solve any Two :**

(A) Define a class Student with attributes name, rollNumber and grade. Write a method display () to display the student's information. 4(CO1)

(B) Elaborate the concepts of java garbage collection mechanism with example. 4(CO1)

(C) Explain the role of constructors in Java programs by analyzing their function in initializing objects, assessing their significance in facilitating object instantiation and evaluating their impact on the overall design and structure of object-oriented programming. 4(CO1)

2. **Solve any Two :**

(A) Implement a superclass Person with fields name and age and a method display (). Then create a subclass Employee that extends person and add field salary and overrides the display () method. Show the use of super to call the superclass method from the subclass. 4.5(CO2)

(B) Illustrate the concept of multiple inheritances in java. Does java support it ? Justify with Example. 4.5(CO2)

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(C) Write a java program to create an interface playable with a method play(). Create a three classes Football, Volleyball and Basketball that implements playable interface and override play () method. 4.5(CO2)

3. Solve any Two :

(A) Implement a class in java to input n number of elements in an array and find the particular elements in the array using linear search. 4(CO3)

(B) Develop a Java program using TreeSet to store product names in alphabetical order and display the sorted product names. 4(CO3)

(C) Create a program to reverse each word in a sentence while preserving the word order. **Analyze** how the choice of data structure affects the performance. 4(CO3)

4. Solve any Two :

(A) Differentiate between final, finally and finalize (). Write a java program to create a method that takes an integer as a parameter and throws an exception if the number is odd. 4.5(CO3)

(B) Design a class that extends Thread and overrides the run () method to print numbers from 1 to 100. Instantiate and start the thread. 4.5(CO3)

(C) Create a user defined exception named check Argument to check the number of argument passed through command line. If number of argument is less than 5 throw the check argument exception else print addition of all 5 numbers. 4.5(CO3)

5. Solve any Two :

(A) Create a program that reads from a character input stream and writes to a character output stream, demonstrating the difference between byte and character streams. 4(CO3)

(B) Write a java program to copy the content of the file "file1.txt" into new file "file2.txt" using character by character. 4(CO3)

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(C) Design a class in java which read and write the content of file at 100th position use the concept of random access file. 4(CO3)

6. Solve any **Two** :

(A) Given a requirement to create a form for user registration, identify the appropriate Swing components (e.g, JLabel, JTextField, JButton) to implement the form with coding example. 4(CO4)

(B) Design a simple application that displays a welcome message using JFrame. What methods would you use to set the title, size and visibility of the window ? 4(CO4)

(C) Design an applet that accepts a number from the user and displays the addition of that number with a predefined value. The applet should include an input text field to accept the number from the user and a button to trigger the calculation of the sum. Upon clicking the button, an event should be generated to perform the addition. 4(CO4)

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B. Tech. Third Semester (Information Technology / CSE / AIDS / AIML) / (SOE_23_IT-101) Examination

Course Code : 23 GE 1303 **Course Name : Linear Algebra**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number.

(2) Write the accurate question number in left margin of answer book along with answers.

(3) All questions carry marks as indicated.

(4) Assume suitable data wherever necessary.

1. Solve any Two :

(A) Solve the system of equations by determinant Method :

$2x + y - z = 3,\ x + y + z = 1,\ x - 2y - 3z = 4.$ 4(CO1)

(B) Solve the system of equations by adjoint Method :

$x + y + z = 3,\ x + 2y + 3z = 4,\ x + 4y + 9z = 6.$ 4(CO1)

(C) If $A = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$, $B = \begin{bmatrix} 5 & -2 \\ 4 & 7 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 \\ 6 & 3 \end{bmatrix}$

and $2A = 3B - 2C$, then find the values of x, y, z and w. 4(CO1)

2. Solve any Two :

(A) Solve the equations by Gauss Elimination Method :

$2x + 3y + z = 13,\ x - y - 2z = -1,\ 3x + y + 4z = 15.$ 4.5(CO1)

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(B) For what value of K, the equation $x + y + z = 1$, $2x + y - 4z = K$, $4x + y + 10z = K^2$ have unique solution and solve them for $k = 1$. 4.5(CO1)

(C) Solve by Crout's Method :

$3x + 2y + 7z = 4$, $2x + 3y + z = 5$, $3x + 4y + z = 7$. 4.5(CO1)

3. Solve any Two :

(A) Express $A^6 - 4A^5 + 8A^4 - 12A^3 + 14A^2$ as a linear polynomial of A, where

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}.$$ 4(CO2)

(B) Determine the eigen value and eigen vector of the matrix :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & 6 \\ 0 & 0 & -3 \end{bmatrix}$$ 4(CO2)

(C) Using Sylvester's theorem show that if $A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}$.

Then $\sin^2 A + \cos^2 A = I$. 4(CO2)

4. Solve any Two :

(A) Let $V = \{(x, y, z) / x, y, z \in R\}$, where R is the field of real numbers then show that $W = \{(x, y, z) / x - 4y + z = 0\}$ is a subspace of vector space V over field R. 4(CO3)

(B) Show that (1, 7, –4) as linear combination of the vectors (1, –3, 2) and (2, –1, 1) in the vector space V_3 of real numbers R. 4(CO3)

(C) Show that the vectors (2, –1, 0), (3, 5, 1), (1, 1, 2) is a basis for R^3. 4(CO3)

5. Solve any Two :

(A) Verify that the following mapping is linear transformations or not ?

$T : V_1 \rightarrow V_3$ defined by $T(x) = (x, x^2, x^3)$. 4(CO4)

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(B) Let $T : V_3 \to V_3$ be a linear map defined by $T(x_1, x_2, x_3) = (x_1, x_2, 0)$. Determine the range and kernel of T. 4(CO4)

(C) Show that the linear map $T : V_3 \to V_3$ defined by $T(x_1, x_2, x_3) = (x_1 + x_2 + x_3, x_2 + x_3, x_3)$ is invertible and find its inverse. 4(CO4)

6. Solve any Two :

(A) Suppose V is a vector space of all polynomials over the unit vector $0 \leq t \leq 1$. If $f(t)$, $g(t) \in V$, inner product on V is defined by $\langle f(t), g(t) \rangle = \int_0^1 f(t) \cdot g(t)dt$. Determine $\langle f, g \rangle$ and $\|g\|$ if $f(t) = t^2 + t - 4$, $g(t) = t - 1$. 4.5(CO4)

(B) Show that the set of vectors : $\{(1, 2, 2), (2, -2, 1), (2, 1, -2)\}$ is orthogonal in R^3. 4.5(CO4)

(C) Use Gram Schmidth orthogonalisation process to convert the basis of R^3 generated by $u_1 = (1, 1, 1)$, $u_2 = (-1, 1, 0)$, $u_3 = (1, 2, 1)$ into an orthogonal basis (V_1, V_2, V_3). 4.5(CO4)

4A) $u = (x_1, x_2, x_3)$

$v = (y_1, y_2, y_3)$ α, β be any scaler

$\alpha u + \beta v = \alpha(x_1 + x_2 + x_3) + \beta(y_1 + y_2 + y_3)$

$= \alpha(x - 4y + z) + \beta(x - 4y + z)$

$= \alpha(0) + \beta(0)$

$= 0$

Hence W is subspace of vector V over field R

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